

Every prior, same slate

The held-out scores that decided which prior is served, then how each one would have ranked today's hitters.

Served (full season)

11
priors compared

32,641
held-out plate appearances

3.05%
held-out HR rate

0.5999
best AUC · KNN comps

served prior

Held-out scores

Fitted on plate appearances before 2026-08-27, scored on the 32,641 that came after (actual HR rate 3.05%). No outcome in the test window was visible during fitting. **Read AUC first** — the slate ranks hitters and takes the top six, so ordering is what matters. Brier and log loss are dominated by the ~3% base rate and barely separate the priors. Metrics are per plate appearance; the probabilities lower down are per game.

PRIOR	AUC	TOP-DECILE LIFT	BRIER	LOG LOSS	CAL. RMSE
Flat league League HR rate as the prior for every hitter.	0.5836	1.54×	0.02952	0.13556	0.00462
KNN comps Prior is the pooled rate of the nearest comparables.	0.5999	1.51×	0.02949	0.13463	0.00494
SVR Support-vector regression on the same feature matrix.	0.5969	1.48×	0.02953	0.13522	0.00618
Random forest Forest regression; a learned similarity metric.	0.5958	1.48×	0.02952	0.13504	0.00551
Core (KNN+SVR) Blend of KNN and SVR, no forest.	0.5999	1.51×	0.02949	0.13463	0.00494
Forest blend KNN + SVR + forest, weights fitted out-of-fold.	0.5999	1.51×	0.02949	0.13463	0.00494
Linear regression Ordinary least squares on the same features.	0.5965	1.48×	0.02951	0.13498	0.00436
Logistic regression Binomial logistic on the per-PA home run outcome.	0.5976	1.52×	0.02953	0.13512	0.00680
Neural net (torch) Two-layer PyTorch MLP, PA-weighted, trained on the same features.	0.5964	1.46×	0.02951	0.13496	0.00542
XGBoost Gradient-boosted trees, depth 3 with subsampling and L2.	0.5971	1.45×	0.02953	0.13503	0.00625

BLEND WEIGHTS (FOREST VARIANT)

KNN 1.00 SVR 0.00 RF 0.00

SHRINKAGE K PER PRIOR

league 248 knn 248 svr 248 rf 248 core 248 ensemble 248
linear 248 logistic 248 nn 248 xgb 248

Estimated by method of moments on out-of-fold residuals, not hand-tuned.

Thin-sample subset

The same held-out window, restricted to hitters with fewer than 100 plate appearances before the cutoff (3,547 PA, actual HR rate 2.34%). For a regular with 600 PA the observed record swamps any prior, so the priors mostly separate here.

PRIOR	AUC	TOP-DECILE LIFT	BRIER	LOG LOSS	CAL. RMSE
Flat league League HR rate as the prior for every hitter.	0.5526	1.44×	0.02288	0.11153	0.00903
KNN comps Prior is the pooled rate of the nearest comparables.	0.5744	0.96×	0.02296	0.11191	0.01517
SVR Support-vector regression on the same feature matrix.	0.5731	1.20×	0.02313	0.11507	0.01888
Random forest Forest regression; a learned similarity metric.	0.5821	1.08×	0.02314	0.11331	0.01866
Core (KNN+SVR) Blend of KNN and SVR, no forest.	0.5744	0.96×	0.02296	0.11191	0.01517
Forest blend KNN + SVR + forest, weights fitted out-of-fold.	0.5744	0.96×	0.02296	0.11191	0.01517
Linear regression Ordinary least squares on the same features.	0.5768	0.60×	0.02307	0.11384	0.01888
Logistic regression Binomial logistic on the per-PA home run outcome.	0.5769	0.60×	0.02314	0.11364	0.02055
Neural net (torch) Two-layer PyTorch MLP, PA-weighted, trained on the same features.	0.5745	1.32×	0.02303	0.11290	0.01770
XGBoost Gradient-boosted trees, depth 3 with subsampling and L2.	0.5886	1.08×	0.02315	0.11327	0.01986

Slate agreement

PRIOR	SHARED	ITS TOP SIX
Served (full season)	6/6	Matt Olson 17.4%, Ronald Acuña Jr. 16.8%, Fernando Tatis Jr. 15.4%, Drake Baldwin 15.2%, Jackson Merrill 14.8%, Teoscar Hernández 14.6%
Flat league	5/6	Ronald Acuña Jr. 16.1%, Matt Olson 15.8%, Fernando Tatis Jr. 15.4%, Drake Baldwin 14.9%, Jackson Chourio 14.1%, Teoscar Hernández 13.9%
KNN comps	6/6	Matt Olson 17.0%, Ronald Acuña Jr. 16.8%, Fernando Tatis Jr. 15.8%, Drake Baldwin 15.4%, Jackson Merrill 14.7%, Teoscar Hernández 14.7%
SVR	6/6	Matt Olson 17.6%, Ronald Acuña Jr. 16.9%, Drake Baldwin 15.6%, Fernando Tatis Jr. 15.6%, Teoscar Hernández 14.8%, Jackson Merrill 14.7%
Random forest	6/6	Matt Olson 17.4%, Ronald Acuña Jr. 16.7%, Drake Baldwin 16.0%, Fernando Tatis Jr. 15.3%, Teoscar Hernández 15.0%, Jackson Merrill 14.8%
Core (KNN+SVR)	6/6	Matt Olson 17.0%, Ronald Acuña Jr. 16.8%, Fernando Tatis Jr. 15.8%, Drake Baldwin 15.4%, Jackson Merrill 14.7%, Teoscar Hernández 14.7%
Forest blend	6/6	Matt Olson 17.0%, Ronald Acuña Jr. 16.8%, Fernando Tatis Jr. 15.8%, Drake Baldwin 15.4%, Jackson Merrill 14.7%, Teoscar Hernández 14.7%
Linear regression	6/6	Matt Olson 17.1%, Ronald Acuña Jr. 16.6%, Fernando Tatis Jr. 15.9%, Drake Baldwin 15.7%, Teoscar Hernández 14.9%, Jackson Merrill 14.8%
Logistic regression	6/6	Matt Olson 17.7%, Ronald Acuña Jr. 16.7%, Fernando Tatis Jr. 16.1%, Drake Baldwin 15.5%, Jackson Merrill 15.1%, Teoscar Hernández 14.9%
Neural net (torch)	6/6	Matt Olson 17.4%, Ronald Acuña Jr. 16.7%, Fernando Tatis Jr. 16.0%, Drake Baldwin 15.4%, Jackson Merrill 14.9%, Teoscar Hernández 14.9%
XGBoost	6/6	Matt Olson 17.4%, Ronald Acuña Jr. 16.6%, Drake Baldwin 16.0%, Fernando Tatis Jr. 15.3%, Teoscar Hernández 15.2%, Jackson Merrill 15.0%

Hitter by hitter

The top 40 bats by the served probability.

ML Model Performance Analysis: Q3 2023											
HITTER		SERVED (FULL SEASON)	FLAT LEAGUE	KNN COMPS	SVR	RANDOM FOREST	CORE (KNN+SVR)	FOREST BLEND	LINEAR REGRESSION	LOGISTIC REGRESSION	NEURAL NET (TO)
1	Matt Olson	slate 17.4%	15.8%	17.0%	17.6%	17.4%	17.0%	17.0%	17.1%	17.7%	
2	Ronald Acuña Jr.	slate 16.8%	16.1%	16.8%	16.9%	16.7%	16.8%	16.8%	16.6%	16.7%	
3	Fernando Tatis Jr.	slate 15.4%	15.4%	15.8%	15.6%	15.3%	15.8%	15.8%	15.9%	16.1%	
4	Drake Baldwin	slate 15.2%	14.9%	15.4%	15.6%	16.0%	15.4%	15.4%	15.7%	15.5%	
5	Jackson Merrill	slate 14.8%	13.5%	14.7%	14.7%	14.8%	14.7%	14.7%	14.8%	15.1%	
6	Teoscar Hernández	slate 14.6%	13.9%	14.7%	14.8%	15.0%	14.7%	14.7%	14.9%	14.9%	
7	Jackson Chourrie	slate 14.1%	14.1%	14.5%	14.6%	14.8%	14.5%	14.5%	14.5%	14.6%	
8	Shohei Ohtani	slate 13.5%	12.1%	13.6%	13.7%	13.8%	13.6%	13.6%	13.5%	13.9%	
9	Will Smith	slate 12.9%	12.7%	12.9%	13.0%	13.6%	12.9%	12.9%	13.6%	13.3%	
10	Manny Machado	slate 12.8%	12.4%	12.8%	13.3%	13.0%	12.8%	12.8%	13.1%	13.2%	
11	Jake Bauers	slate 12.5%	11.6%	12.8%	12.7%	12.7%	12.8%	12.8%	12.5%	12.5%	
12	Mookie Betts	slate 12.5%	12.4%	12.2%	13.0%	12.3%	12.2%	12.2%	12.6%	12.7%	
13	Austin Riley	slate 12.3%	11.4%	12.1%	12.1%	11.9%	12.1%	12.1%	12.1%	12.2%	
14	William Contreras	slate 12.2%	12.0%	12.2%	12.0%	12.0%	12.2%	12.2%	12.1%	12.1%	
15	Michael Harris	slate 11.4%	11.5%	11.3%	11.9%	11.3%	11.3%	11.3%	11.6%	12.0%	
16	Ozzie Albies	slate 11.1%	11.5%	11.1%	11.7%	11.7%	11.1%	11.1%	11.4%	11.3%	
17	Freddie Freeman	slate 10.6%	10.3%	10.3%	10.2%	10.1%	10.3%	10.3%	10.4%	10.2%	
18	Ty France	slate 10.5%	10.3%	10.7%	10.6%	10.8%	10.7%	10.7%	10.7%	10.7%	
19	Andy Pages	slate 10.3%	9.9%	9.9%	9.7%	10.1%	9.9%	9.9%	9.8%	9.7%	
20	Ethan Salas	slate 10.0%	9.7%	9.5%	9.5%	9.4%	9.5%	9.5%	9.5%	9.2%	
21	Brice Turang	slate 9.8%	10.0%	9.5%	9.8%	9.4%	9.5%	9.5%	9.2%	9.1%	
22	Christian Yelich	slate 9.7%	9.9%	10.0%	9.6%	9.8%	10.0%	10.0%	10.1%	9.7%	

HITTER	SERVED (FULL SEASON)	FLAT LEAGUE	KNN COMPS	SVR	RANDOM FOREST	CORE (KNN+SVR)	FOREST BLEND	LINEAR REGRESSION	LOGISTIC REGRESSION	NEURAL NET (TO
23Garrett MitchellMIL	9.2%	8.8%	9.2%	8.0%	8.5%	9.2%	9.2%	8.7%	8.6%	
24Xander BogaertsSD	8.7%	8.8%	8.6%	8.3%	8.5%	8.6%	8.6%	8.7%	8.4%	
25Enrique Hernández	8.3%	8.3%	8.1%	7.5%	7.6%	8.1%	8.1%	7.9%	7.5%	
26Kyle TuckerLAD	8.2%	8.3%	8.3%	8.0%	8.2%	8.3%	8.3%	8.1%	7.9%	
27Miguel RojasLAD	7.1%	8.8%	7.3%	7.6%	7.6%	7.3%	7.3%	7.6%	7.8%	
28Mauricio DubónTL	6.9%	8.3%	7.0%	7.4%	7.2%	7.0%	7.0%	6.8%	7.2%	
29Ha-Seong KimATL	6.0%	7.2%	5.2%	4.6%	4.5%	5.2%	5.2%	5.6%	5.5%	
30Joey OrtizMIL	5.7%	6.6%	5.9%	5.9%	5.9%	5.9%	5.9%	5.8%	5.8%	
31Cooper PrattMIL	5.2%	6.8%	5.2%	4.8%	4.9%	5.2%	5.2%	4.7%	4.7%	
32Dustin HarrisSD	5.0%	6.1%	4.9%	4.2%	4.3%	4.9%	4.9%	4.8%	4.7%	
33Sal FrelickMIL	4.7%	5.8%	4.6%	4.5%	4.4%	4.6%	4.6%	4.5%	4.5%	
34Samad TaylorSD	4.7%	7.2%	5.3%	4.4%	4.8%	5.3%	5.3%	4.7%	5.0%	
35Jake Cronenworth	4.5%	5.7%	4.6%	4.6%	4.7%	4.6%	4.6%	4.7%	4.7%	

Per-game probabilities: each prior's per-PA rate run through this hitter's own plate appearances, park, weather, bullpen and times-through-the-order terms. Only the prior changes across the columns — every other term is held fixed, so a left-to-right spread is the prior disagreeing about the hitter, not the context. Shading grows with the probability.